

# **EXHIBIT 35**

## **EXHIBIT 35**

### **U.S. Patent No. 9,093,425 Infringement Chart**

**Title:** Self-aligned liner formed on metal semiconductor alloy contacts

**Inventors:** Nicolas Breil, Christian Lavoie, Ahmet S. Ozcan, Kathryn T. Schonenberg, Jian Yu

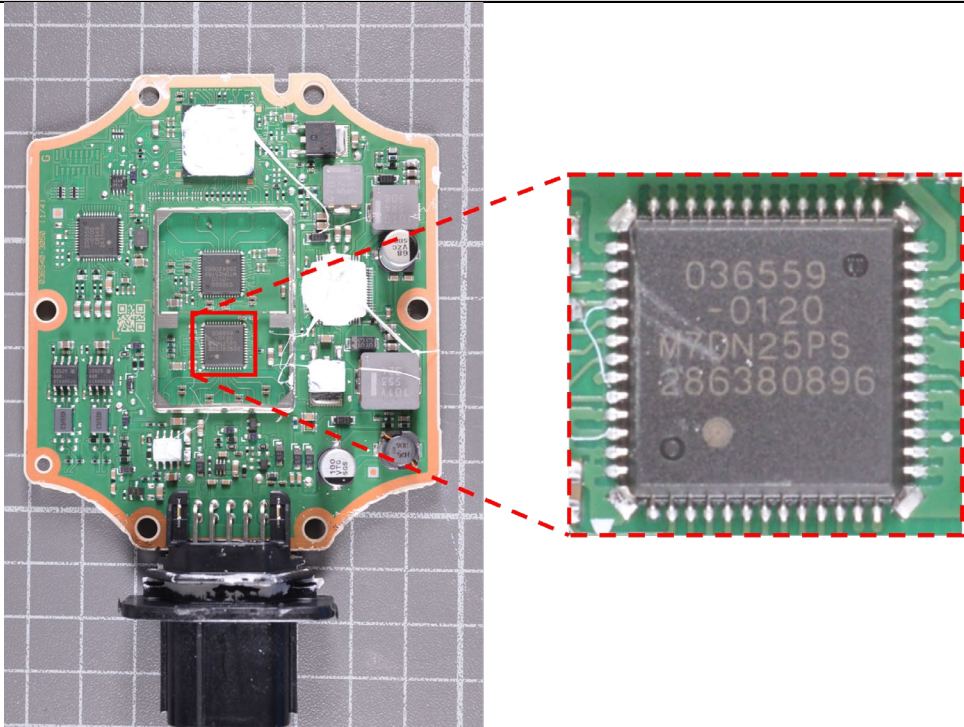
**Abstract:** Metal semiconductor alloy contacts are provided on each of a source region and a drain region which are present in a semiconductor substrate. A transition metal is then deposited on each of the metal semiconductor alloy contacts, and during the deposition of the transition metal, the deposited transition metal reacts preferably, but not necessarily always, in-situ with a portion of each the metal semiconductor alloy contacts forming a transition metal-metal semiconductor alloy liner atop each metal semiconductor alloy contact. Each transition metal-metal semiconductor alloy liner that is provided has outer edges that are vertically coincident with outer edges of each metal semiconductor alloy contact. The transition metal-metal semiconductor alloy liner is more etch resistant as compared to the underlying metal semiconductor alloy. As such, the transition metal-metal semiconductor alloy liner can serve as an effective etch stop layer during any subsequently performed etch process.

| <b>Claim</b>                                 | <b>Tower '425 Accused Products</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| [1pre] A semiconductor structure comprising: | <p>To the extent the preamble is limiting, devices manufactured by Tower that rely on its SiGe and BiCMOS processes (including the Tower SBC18 process) (hereinafter, "Tower '425 Accused Products") comprise a semiconductor structure.<sup>1</sup></p> <p>For example, the Toyota Blind Spot Monitor Sensor (#88162-0R031) includes a module manufactured by DENSO Corporation (identified as DNSRR004), which further includes a die manufactured by Tower (identified as M7DN25PS) that is, on information and belief, representative of devices manufactured by Tower that rely on its SBC18 process.</p> |

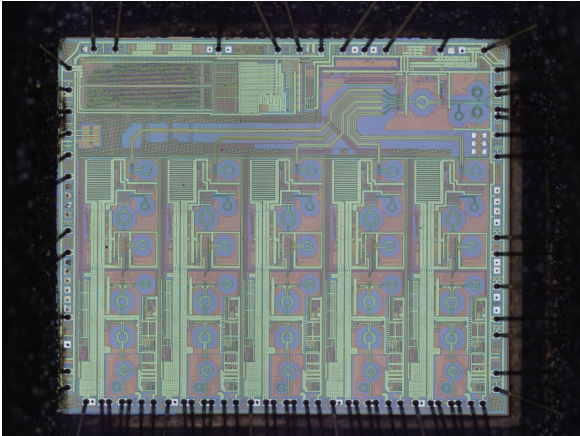
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<sup>1</sup> GlobalFoundries reserves the right to identify to identify different or additional Tower '425 Accused Products and amend or supplement these charts after further investigation and discovery, claim construction, or other meriting circumstances. Such supplemental information may include, but is not limited to, source code, data sheets, design specifications, deposition testimony, testing information, reference designs implementation and utilization information, and/or schematics.

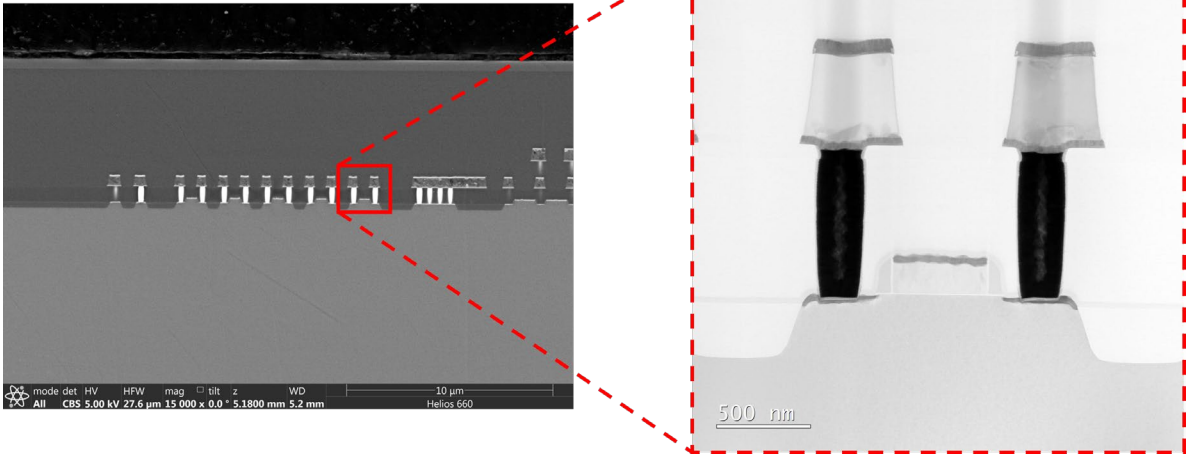
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| Claim | Tower '425 Accused Products                                                        |
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### EXHIBIT 35

| Claim                                                                                                                                                                                                                                                   | Tower '425 Accused Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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|                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>[1a] a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region;</p> | <p>The Tower '425 Accused Products comprise a semiconductor substrate having a source region and a drain region located within a semiconductor material portion of said semiconductor substrate, wherein said source region and said drain region are spaced apart by a channel region.</p> <p>For example, below is a cross-sectional image of a transistor in the Tower '425 Accused Products. The image shows a semiconductor substrate (bottom) having a source region (left) and a drain region (right) that are located within a semiconductor material portion of the semiconductor substrate. The source and drain regions are spaced apart by a channel region (center).</p> |

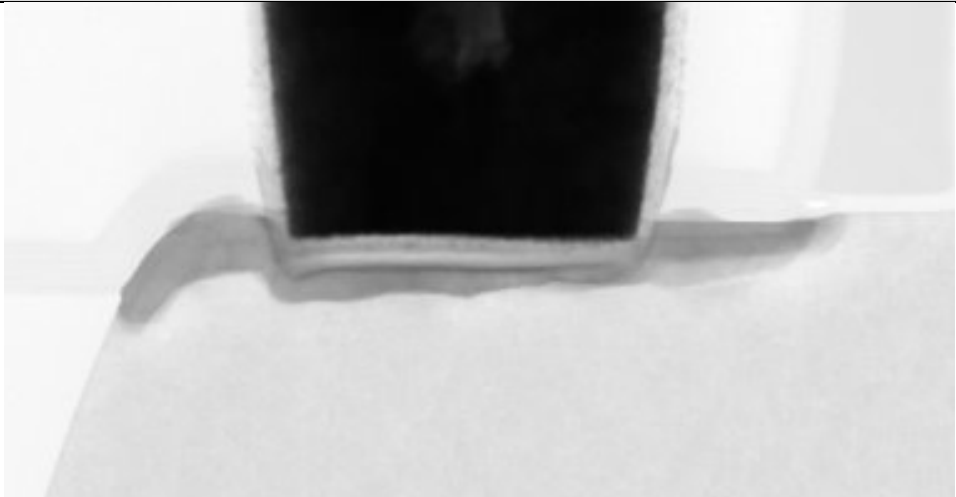
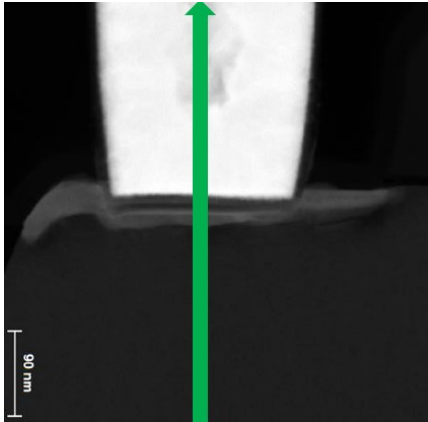
### EXHIBIT 35

| Claim                                                                      | Tower '425 Accused Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
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|                                                                            |  <p>The image is a scanning electron micrograph (SEM) of a device. The main image shows a series of vertical structures on a substrate. A red box highlights a specific region of these structures. A dashed red line connects this box to a magnified inset on the right. The inset shows two vertical structures with a central gap, sitting on a base. A scale bar in the inset indicates 500 nm. The main image has a scale bar indicating 10 μm. Technical data at the bottom of the main image includes: mode det HV HFW mag tilt z WD 10 μm Helios 660.</p> |
| <p>[1b] a functional gate structure located above said channel region;</p> | <p>The Tower '425 Accused Products comprise a functional gate structure located above said channel region. For example, the Tower '425 Accused Products includes a functional gate structure (middle) that is located above the channel region. <i>See</i> limitation [1a].</p>                                                                                                                                                                                                                                                                                                                                                                      |

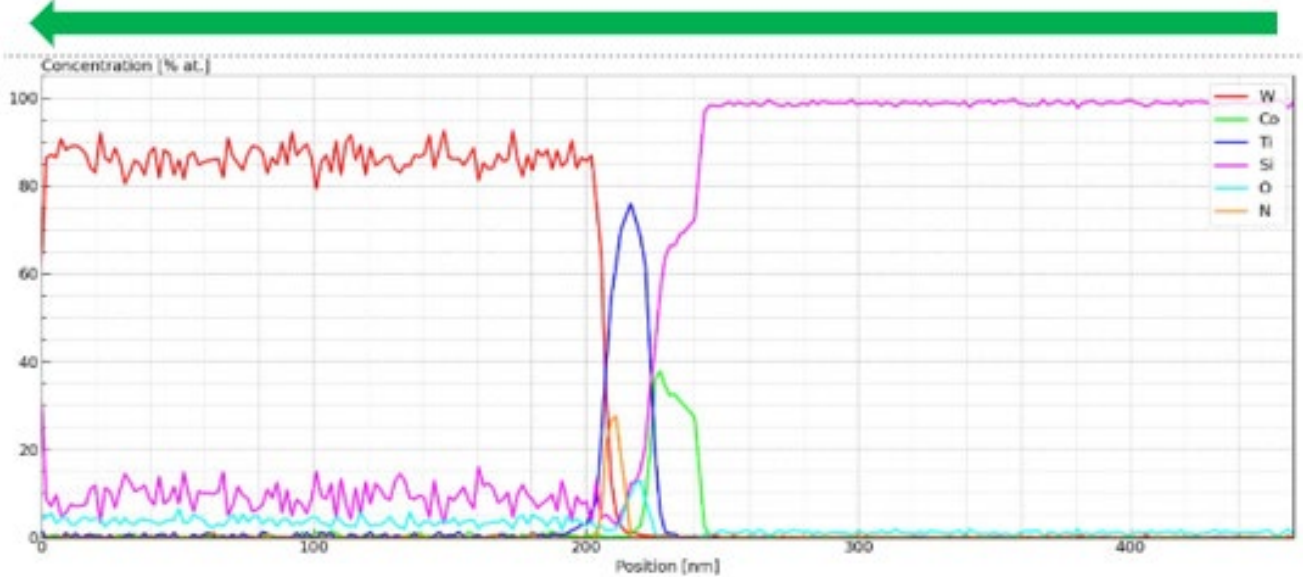
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| Claim                                                                                                                                                                                                                                                                                | Tower '425 Accused Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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|                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>[1c] a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region; and</p> | <p>The Tower '425 Accused Products comprise a source-side material stack of, from bottom to top, a source-side metal semiconductor alloy portion and a source-side transition-metal metal semiconductor alloy portion located on one side of said functional gate structure and located atop said source region.</p> <p>For example, the image below shows a close-up of the source-side material stack in the Tower '425 Accused Products (i.e., the left side of the image immediately above in limitation [1b]).</p> |

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| Claim | Tower '425 Accused Products                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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|       | <div data-bbox="751 228 1701 722"></div> <p data-bbox="562 764 1892 943">Analysis of the source-side material stack shows the presence of a metal semiconductor alloy portion located beneath a transition-metal metal semiconductor alloy portion. For example, the green arrow in the image below (i.e., from bottom to top) corresponds to the right-to-left direction in the ensuing graph. As shown in the chart below, the material stack comprises cobalt silicide and titanium and/or tungsten, cobalt, silicon alloy.</p> <div data-bbox="1014 979 1438 1398"></div> |

## EXHIBIT 35

| Claim                                                                                                                                                                                                                                                                            | Tower '425 Accused Products                                                                                                                                                                                                                                                                                                                                                                                                            |
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| <p>[1d] a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.</p> | <p>The Tower '425 Accused Products comprise a drain-side material stack of, from bottom to top, a drain-side metal semiconductor alloy portion and a drain-side transition metal-metal semiconductor alloy portion located on another side of said functional gate structure and located atop said drain region.</p> <p>The drain-side material stack has the same composition as the source-side material stack. <i>See</i> [1c].</p> |